

Kinematics

§1 Trajectories

time-dependent
differential eqns $\{x, y, z\}$

Streamlines

$$\underline{v} \parallel d\underline{s}$$

$$\frac{dx}{v_x} = \frac{dy}{v_y} = \frac{dz}{v_z}$$

Steady flow ($\frac{\partial \underline{v}}{\partial t} = 0$) traj. = streaml.

Eulerian: variations at all fixed positions

Lagrangian: follow fluid elements moving with the flow.

Time derivative (Material) $f = f(t, \underline{r})$

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \frac{\partial}{\partial x} \frac{dx}{dt} + \frac{\partial}{\partial y} \frac{dy}{dt} + \frac{\partial}{\partial z} \frac{dz}{dt}$$

$$= \frac{\partial}{\partial t} + \underline{v} \cdot \underline{\nabla}$$

↓
local

↓
Convective
transport

Along a streamline in a steady flow,

$$\frac{DQ}{Dt} = \underline{v} \cdot \underline{\nabla} Q = |\underline{v}| \hat{s} \cdot \underline{\nabla} Q = |\underline{v}| \frac{\partial Q}{\partial s}$$

$$\text{If } \underline{v} \cdot \underline{\nabla} Q = 0$$

implies that $\frac{\partial Q}{\partial s} = 0$

$Q = \text{const. along streamline.}$

Deformation tensor

$$D_{ij} = \frac{\partial v_i}{\partial x_j}$$

$$\text{where } dv_i = \underbrace{\left(\frac{\partial v_i}{\partial x_j} \right)}_{D_{ij}} dx_j$$

$$D_{ij} = e_{ij} + \omega_{ij}$$

$$e_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) \quad \begin{array}{l} \text{(symmetric)} \\ \text{rate of strain tensor} \end{array}$$

$$\omega_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i} \right) \quad \text{vorticity tensor}$$

Generalised deformation tensor:

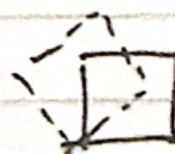
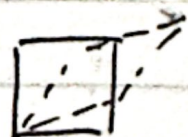
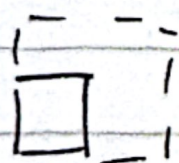
$$D_{ij} = \underbrace{t_{ij}} + \underbrace{d_{ij}}_{e_{ij}} + \omega_{ij}$$

↓
diagonal,
rate of
volume
expansion.

↓
symmetric,
off-diagonal.
'Shearing'
deformation
without
changing
volume.

↓
antisymmetric,
local rigid body
rotation.

$$\underline{\omega} = \nabla \times \underline{v}$$



$$\begin{cases} t_{ij} = \frac{1}{3} \delta_{ij} e_{kk} \\ d_{ij} = e_{ij} - \frac{1}{3} \delta_{ij} e_{kk} \end{cases}$$

(trace)

(0 = trace)

Mass conservation

$$EL: \frac{d}{dt} \iiint_V \rho dV = - \oint_S \rho \underline{v} \cdot d\underline{S}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \underline{v}) = 0$$

↓

$$\begin{aligned} \frac{D\rho}{Dt} &= \frac{\partial \rho}{\partial t} + (\underline{v} \cdot \nabla) \rho = -\nabla \cdot (\rho \underline{v}) + (\underline{v} \cdot \nabla) \rho \\ &= -\rho (\nabla \cdot \underline{v}) \end{aligned}$$

$$\frac{D\rho}{Dt} + \rho (\nabla \cdot \underline{v}) = 0$$

$$La: \frac{d}{dt} \iiint_V \rho(t) dx(t) dy(t) dz(t) = 0$$

$$\iiint_V \left(\frac{d\rho}{dt} + \rho \left(\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} \right) \right) dx dy dz = 0$$

↓

$$\frac{D\rho}{Dt} + \rho (\nabla \cdot \underline{v}) = 0$$

$$\begin{aligned} \underline{\text{Incompressibility}} &\rightarrow \nabla \cdot \underline{v} = 0 \\ &\text{or } \frac{D\rho}{Dt} = 0 \end{aligned}$$

$$\underline{\text{Irrotationality}} \quad \nabla \times \underline{v} = \underline{\omega} = 0$$

$$\rightarrow \underline{v} = \nabla \phi \quad \rightarrow \text{velocity potential}$$

if irrotational & incompressible

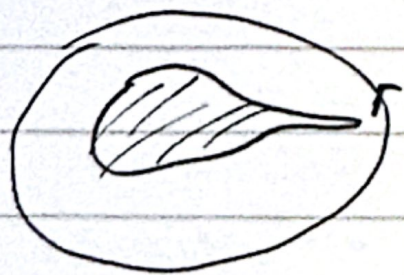
$$\nabla \cdot \underline{v} = \nabla \cdot (\nabla \phi) = 0 \rightarrow \underline{\nabla^2 \phi = 0}$$

Laplace's eqn. $\nearrow$ (potential flow)

Circulation $\Gamma = \oint_C \underline{v} \cdot d\underline{l} = \iint_S (\nabla \times \underline{v}) \cdot d\underline{S}$
 $= \iint_S \underline{\omega} \cdot d\underline{S}$

for irrotational flow, $\underline{\omega} = 0 \rightarrow \Gamma = 0$,

Circulation is the same around all simply closed contours enclosing the aerofoil.



C (any such, Cauchy's theorem)

Stream function

(incompressible fluid)

$$\nabla \cdot \underline{v} = 0$$

$$\hookrightarrow \underline{v} = \nabla \times \underline{A}$$

In 2D $\rightarrow$ pick $A_z = \psi$ (stream function)

$$\begin{cases} v_x = \frac{\partial \psi}{\partial y} \\ v_y = -\frac{\partial \psi}{\partial x} \end{cases}$$

(Cartesian)

$$\begin{cases} v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} \\ v_\theta = -\frac{\partial \psi}{\partial r} \end{cases}$$

(Polar)

2nd order $\rightarrow \nabla^2 \psi = 0$ (Laplace's eqn)

$$\underline{v} \cdot \underline{\nabla} \psi = v_x \frac{\partial \psi}{\partial x} + v_y \frac{\partial \psi}{\partial y} = v_x (-v_y) + v_y (v_x) = 0$$

ψ is const along streamlines.

$$\left(\frac{\partial \psi}{\partial s} = 0 \right)$$

$$\underline{\nabla} \phi \cdot \underline{\nabla} \psi = 0$$

(equipotentials $\perp$ streamlines)

$$\phi = \text{const.}$$

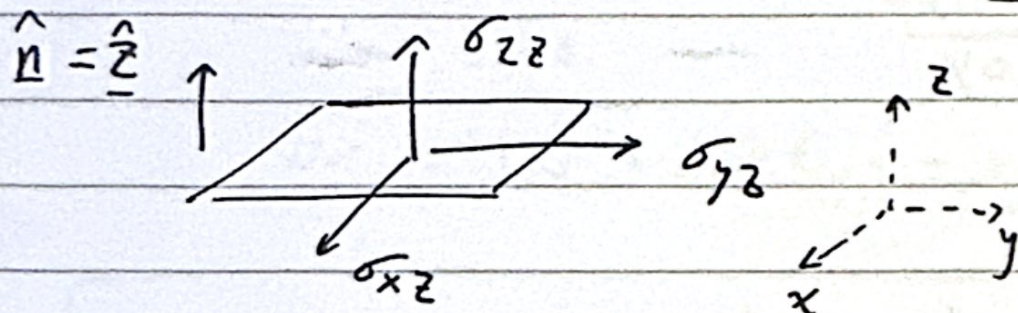
(equipotentials)

$$\psi = \text{const.}$$

(streamlines)

§ 2 Dynamics

Stress Tensors



local stress
(force/area)

$$\underline{T_i} = \underline{\sigma_{ij}} n_j$$

(i-th direction)

σ_{ij} : i-th component of stress tensor on a surface element with a normal n_j in j-th direction.

if $\begin{cases} i \neq j \\ i = j \end{cases}$ σ_{ij} tangential
 σ_{ij} normal to plane

Shear flow in a gas

$$\delta^2 P_x = 2 \underbrace{\left(\frac{1}{6} n u \delta t \delta S \right)}_{(\#) \delta N} \underbrace{\left(m \lambda \frac{dv_x}{dy} \right)}_{\delta P \text{ (1 particle)}}$$

$$\delta F_x = \frac{1}{3} n m u \lambda \frac{dv_x}{dy} \delta S$$

$$T_x = \frac{\delta F_x}{\delta S} = \sigma_{yx} = \underbrace{\frac{1}{3} n m u \lambda \frac{dv_x}{dy}}_{\eta}$$

(dynamic viscosity)

$$\nu = \frac{\eta}{\rho} \quad (\text{Kinematic viscosity})$$

Surface forces

$$F_{\text{visc}} = f_{\text{visc},x} \delta S \delta y$$

$$f_{\text{visc},x} = \frac{d\sigma_{xy}}{dy}$$

viscous force, per unit vol.

Generalised stress tensor

(Incompressible, Newtonian fluid)

$$\sigma_{ij} = \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) = 2\eta e_{ij}$$

(symmetric)

rate of strain.

$$f_{\text{visc},i} = \frac{d\sigma_{ij}}{dx_j}$$

Total stress (include pressure)

$$\Sigma_{ij} = \sigma_{ij} - p \delta_{ij}$$

N-S Eqn

$$\frac{\partial}{\partial t} (\rho v_i) + \nabla \cdot (\rho v_i \underline{v}) = - \frac{\partial p}{\partial x_i} + \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i$$

↓
flow
transport

↓
pressure

↓
viscous

↓
external

$$\underbrace{V_i \left(\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \underline{v}) \right)}_0 + \rho \frac{\partial v_i}{\partial t} + \rho (\underline{v} \cdot \nabla) v_i = \text{RHS}$$

Mass Cons.

Note also that $\frac{\partial \sigma_{ij}}{\partial x_j} = \eta \nabla^2 v_i = \rho \nu \nabla^2 v_i$

$$\left[\frac{\partial \underline{v}}{\partial t} + (\underline{v} \cdot \nabla) \underline{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \underline{v} + \underline{g} \right]$$

NS Eqn. for incompressible, Newtonian fluid.

$$Re = \frac{\text{inertial}}{\text{viscous}} = \frac{|(\underline{v} \cdot \nabla) \underline{v}|}{|\nu \nabla^2 \underline{v}|} = \frac{U^2/L}{\nu U/L^2} = \frac{UL}{\nu}$$

Euler equation ($\nu = 0$)

~~$$\rho \frac{\partial \underline{v}}{\partial t} + (\underline{v} \cdot \nabla) \underline{v} = -\frac{1}{\rho} \nabla p + \underline{g}$$~~

$$\rho \frac{\partial \underline{v}}{\partial t} + \rho (\underline{v} \cdot \nabla) \underline{v} = -\nabla p + \rho \underline{g}$$

valid even if compressible

ideal: inviscid & incompressible

Boundary Conditions

① rigid boundary

$$(V_{\text{fluid}})_\perp = (V_{\text{boundary}})_\perp \quad (\text{no penetration})$$

$$(V_{\text{fluid}})_\parallel = (V_{\text{boundary}})_\parallel \quad (\text{no slip,} \\ \underline{\text{only for viscous}})$$

• For viscous fluid, $\underline{V}_{\text{fluid}} = \underline{V}_{\text{boundary}}$

② Interface (btw. 2 fluids)

incompressible fluid:

$$\iiint_V \nabla \cdot \underline{v} \, dV = 0$$

$$\oint_S \underline{v} \cdot d\underline{S} = 0 \rightarrow \underline{(v_2 - v_1) \cdot \hat{n}} = 0$$

(Kinematic
boundary condition)

if interface is

given by $z = f(x, y, t)$

$$\underline{v_{i,z} = \frac{dz}{dt} = \frac{\partial f}{\partial t} + v_{i,x} \frac{\partial f}{\partial x} + v_{i,y} \frac{\partial f}{\partial y}}$$

($i=1, 2$, fluid label)

also K. b.c.

viscous fluid:

$$\underline{V}_{\text{fluid 1}} = \underline{V}_{\text{fluid 2}} = \underline{V}_{\text{interface}}$$

Surface tension

γ : work per unit area that has to be done to increase the surface.
(molecules oppose increase of surface area)

$$dW = \gamma dS = F d\ell$$

$\underbrace{\quad}_{L d\ell}$

$$\underline{F = \gamma L}$$

Continuity of the stress: ($f_{\text{surf}, i}$ negligible)

$$\underbrace{((\sigma_{ij})^{(2)} - p^{(2)} \delta_{ij})}_{\Sigma_{ij}} n_j = ((\sigma_{ij})^{(1)} - p^{(1)} \delta_{ij}) n_j$$

Free surface (liquid & gas)

$$(1) \quad V_z = \frac{dz}{dt} = \frac{\partial f}{\partial t} + v_x \frac{\partial f}{\partial x} + v_y \frac{\partial f}{\partial y}$$

kinematic b.c.

(2) Continuity of the stress still holds.
When both fluid & gas inviscid,
at the free surface:

• Net pressure force is equal & opposite
to the normal component of surface tension.
(later chap. detail)

Vorticity eqn

ρ const & fluid incompressible

$$\underline{g} = -\nabla \chi$$

$$\frac{\partial \underline{v}}{\partial t} + (\underline{v} \cdot \nabla) \underline{v} = -\nabla \left(\frac{p}{\rho} + \chi \right) + \nu \nabla^2 \underline{v}$$

$$\text{since } (\underline{v} \cdot \nabla) \underline{v} = \underbrace{(\nabla \times \underline{v}) \times \underline{v}}_{\underline{\omega} \times \underline{v}} + \frac{1}{2} \nabla v^2$$

$$\frac{\partial \underline{v}}{\partial t} + \underline{\omega} \times \underline{v} = -\nabla \left(\frac{1}{2} v^2 + \frac{p}{\rho} + \chi \right) + \nu \nabla^2 \underline{v}$$

take $\nabla \times (\dots)$

$$\frac{\partial \underline{\omega}}{\partial t} + \nabla \times (\underline{\omega} \times \underline{v}) = \nabla \times (\nu \nabla^2 \underline{v})$$

$$\frac{\partial \underline{\omega}}{\partial t} + (\underline{v} \cdot \nabla) \underline{\omega} - \underbrace{(\underline{\omega} \cdot \nabla) \underline{v}}_0 + \underbrace{\underline{\omega} (\nabla \cdot \underline{v})}_{0} - \underbrace{\underline{v} (\nabla \cdot \underline{\omega})}_{0}$$

$$= \nabla \times (\nu \nabla^2 \underline{v}) = \nu \nabla^2 \underline{\omega}$$

$$\boxed{\frac{\partial \underline{\omega}}{\partial t} + (\underline{v} \cdot \nabla) \underline{\omega} = (\underline{\omega} \cdot \nabla) \underline{v} + \nu \nabla^2 \underline{\omega}}$$

$\frac{D\underline{\omega}}{Dt}$

rate of
change of
vorticity

vortex
stretching

molecular
diffusion
of vorticity

Ideal fluid & Kelvin's Theorem

Inviscid ($\nu=0$)

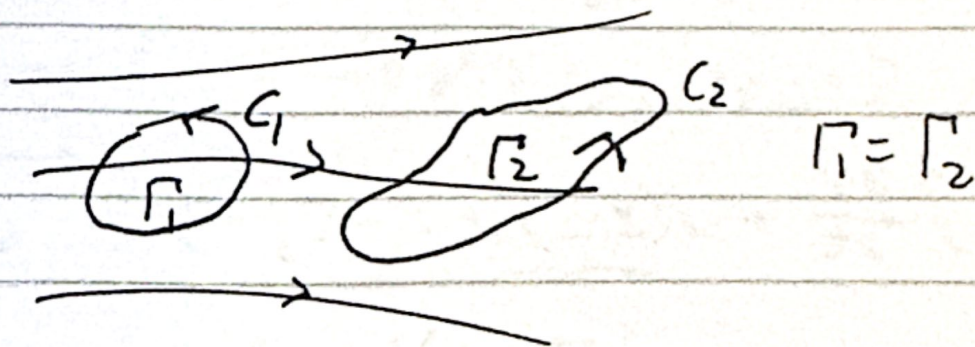
velocity eqn $\rightarrow \frac{D\mathbf{u}}{Dt} = (\mathbf{u} \cdot \nabla) \mathbf{u}$

Circulation $\Gamma = \oint_C \mathbf{u} \cdot d\mathbf{l}$

$$\begin{aligned} \frac{D\Gamma}{Dt} &= \oint_C \frac{D\mathbf{u}}{Dt} \cdot d\mathbf{l} + \oint_C \mathbf{u} \cdot \frac{D(d\mathbf{l})}{Dt} \\ &= \underbrace{-\oint_C \nabla \left(\frac{p}{\rho} + \chi \right) \cdot d\mathbf{l}}_0 + \underbrace{\oint_C \nabla \left(\frac{1}{2} \mathbf{u}^2 \right) \cdot d\mathbf{l}}_0 \end{aligned}$$

$$\frac{D}{Dt} \left(\oint_C \mathbf{u} \cdot d\mathbf{l} \right) = 0$$

- circulation around a closed contour is constant, as the contour moves with the flow.



$$\text{or } \frac{D\Gamma}{Dt} = \frac{D}{Dt} \left(\iint_S \underbrace{(\nabla \times \mathbf{u})}_{\boldsymbol{\omega}} \cdot d\mathbf{S} \right) = 0$$

Helmholtz's Theorem

A vortex line attached to a fluid element remains attached as it moves with the flow.

2d-flow $\underline{v} = v_x \hat{x} + v_y \hat{y}$

$$(\underline{\omega} \cdot \underline{\nabla}) \underline{v} = \omega \frac{\partial \underline{v}}{\partial z} = 0$$

$\hat{z}$ direction

$$\frac{D\omega}{Dt} = 0 \text{ for 2D ideal flow}$$

(based on vorticity eqn)

• steady flow

$$\frac{\partial \omega}{\partial t} = 0 \rightarrow (\underline{v} \cdot \underline{\nabla}) \omega = 0$$

vorticity
conserved
along a
streamline.

Bernoulli's theorem

ideal fluid,

• Euler's eqn:

$$\rho \frac{\partial \underline{v}}{\partial t} + \rho (\underline{v} \cdot \underline{\nabla}) \underline{v} = -\underline{\nabla} p + \rho \overbrace{(-\underline{\nabla} \chi)}^g$$

$$\frac{1}{2} \underline{\nabla} (\underline{v} \cdot \underline{v}) - \underline{v} \times (\underline{\nabla} \times \underline{v})$$

$$\rho \frac{\partial \underline{v}}{\partial t} + \rho \underline{\omega} \times \underline{v} = -\underline{\nabla} \left(\frac{1}{2} \rho v^2 + p + \rho \chi \right)$$

Steady flow $\rightarrow \frac{\partial \underline{v}}{\partial t} = 0$

Pot with $\underline{v} \cdot (\dots)$

$$(\underline{v} \cdot \nabla) \left(\frac{1}{2} \rho v^2 + p + \rho \chi \right) = 0$$

$$H = \underbrace{\frac{1}{2} \rho v^2 + p}_{\text{dynamic pressure}} + \rho \chi = \text{const.} \quad (\text{along a streamline})$$

Bernoulli's theorem
for steady & ideal fluid
(const. ρ)

$(\frac{1}{2} \rho v^2 + p)$: total pressure

Irrotational flow $\nabla \times \underline{v} = 0, \quad \underline{v} = \nabla \phi$

$$\rho \frac{\partial \underline{v}}{\partial t} + \underbrace{\rho \underline{\omega} \times \underline{v}}_0 = -\nabla \left(\frac{1}{2} \rho v^2 + p + \rho \chi \right)$$

$$\nabla \left(\rho \frac{\partial \phi}{\partial t} \right) = -\nabla \left(\frac{1}{2} \rho v^2 + p + \rho \chi \right)$$

$$\rho \frac{\partial \phi}{\partial t} + \frac{1}{2} \rho v^2 + p + \rho \chi = \text{const.}$$

Subsume const. into ϕ
 $\downarrow$

$$\rho \frac{\partial \phi}{\partial t} + \underbrace{\frac{1}{2} \rho v^2 + p + \rho \chi}_0 = 0$$

Constant throughout the fluid!

§ 3 Potential Flows

Assumptions

inviscid $\nu = 0$

irrotational $\nabla \times \underline{V} = 0 \rightarrow \underline{V} = \nabla \phi$

incompressible $\nabla \cdot \underline{V} = 0 \rightarrow \nabla \cdot \underline{V} = \boxed{\nabla^2 \phi = 0}$

Laplace's eqn

b.c.

rigid boundary

$$\left. \frac{\partial \phi}{\partial n} \right|_{\text{surf}} = V_n, \text{ boundary}$$

interface

$$\left. \frac{\partial \phi}{\partial n} \right|_{\text{fluid 1, surf}} = \left. \frac{\partial \phi}{\partial n} \right|_{\text{fluid 2, surf}}$$

no slip ~~Not~~ applicable (inviscid)

Euler's eqn

$$\rho \frac{\partial \underline{V}}{\partial t} + \rho (\underline{V} \cdot \nabla) \underline{V} = -\nabla p + \rho \underline{g}$$

Bernoulli's theorem (irrotational)

$$\rho \frac{\partial \phi}{\partial t} + \frac{1}{2} \rho V^2 + p + \rho \chi = \text{const.}$$

Can define streamline function ψ

for 2d flow

$$V_x = \frac{\partial \psi}{\partial y}$$

$$V_y = -\frac{\partial \psi}{\partial x}$$

or polar coords: $V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}$ $V_\theta = -\frac{\partial \psi}{\partial r}$

$\psi = \text{const.} \rightarrow \text{streamlines}$

$\phi = \text{const.} \rightarrow \text{equipotentials}$

$$(\nabla \times \underline{V})_z = -\nabla^2 \psi = 0 \rightarrow \psi \text{ satisfies Laplace's eqn.}$$

Canonical Examples

① Linear potential flow

$$\underline{V} = U \hat{x}$$

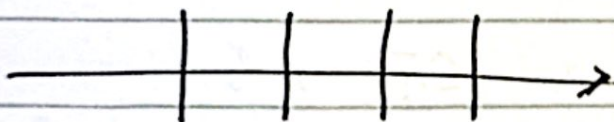
$$\omega = 0 \quad \checkmark$$

$$\underline{V} = \nabla \phi \rightarrow \underline{\phi} = Ux$$

$$2d: \quad \frac{\partial^2 \psi}{\partial y^2} = V_x = U \rightarrow \underline{\psi} = Uy$$

$\psi = \text{const} \rightarrow y = \text{const} \quad \underline{\text{streamlines}}$

$\phi = \text{const} \rightarrow x = \text{const} \quad \underline{\text{equipotentials}}$



② Line vortex flow

$$2d \text{ polar} \quad \underline{V} = \frac{\Gamma}{2\pi r} \hat{\theta}$$

$$\omega = \nabla \times \underline{V} = \frac{1}{r} \frac{\partial}{\partial r} (r V_\theta) \hat{z} = \underline{0}$$

$$\frac{1}{r} \frac{\partial \phi}{\partial \theta} = V_{\theta} = \frac{\Gamma}{2\pi r} \rightarrow \underline{\underline{\phi = \frac{\Gamma \theta}{2\pi}}}$$

obtain ψ by $V_{\theta} = -\frac{\partial \psi}{\partial r} \Rightarrow \underline{\underline{\psi = -\frac{\Gamma}{2\pi} \ln\left(\frac{r}{r_0}\right)}}$

• Vorticity is non-zero along a line.

Sources/Sinks

$$\underline{V} = \frac{m}{4\pi r^2} \hat{r} \quad (3D)$$

$$\text{or } \underline{V} = \frac{m}{2\pi r} \hat{r} \quad (2D)$$

$$\underline{\omega} = \nabla \times \underline{V} = 0$$

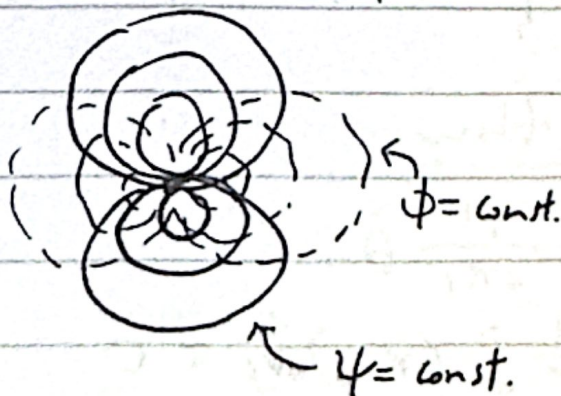
$$2D \rightarrow \frac{\partial \phi}{\partial r} = \frac{m}{2\pi r} \rightarrow \underline{\underline{\phi = \frac{m}{2\pi} \ln\left(\frac{r}{r_0}\right)}}$$

$$\frac{d\psi}{d\theta} = \frac{m}{2\pi} \rightarrow \underline{\underline{\psi = \frac{m\theta}{2\pi}}}$$

(opposite of line vortex)

Dipole flow

(see C.T. notes)



Complex Potential (★)

$$u = \frac{\partial \phi}{\partial x} \quad v = \frac{\partial \phi}{\partial y}$$
$$= \frac{\partial \psi}{\partial y} \quad = -\frac{\partial \psi}{\partial x}$$

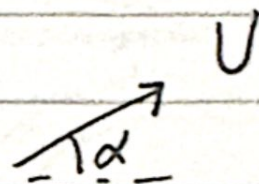
Def $\omega = \phi + i\psi$

$$\rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad \& \quad \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\frac{d\omega}{dz} = \frac{\partial \phi}{\partial x} + i \frac{\partial \psi}{\partial x} = \boxed{u - iv}$$

flow speed $q = \sqrt{u^2 + v^2} = \left| \frac{d\omega}{dz} \right|$

Uniform flow at angle α



$$u = U \cos \alpha$$

$$v = U \sin \alpha$$

$$\frac{d\omega}{dz} = U e^{-i\alpha}$$

$$\underline{\omega = U z e^{-i\alpha}}$$

Line Vortex

$$u = \frac{\Gamma}{2\pi r} \hat{\theta} \quad \phi = \frac{\Gamma \theta}{2\pi}, \quad \psi = -\frac{\Gamma}{2\pi} \ln r$$

$$\omega = \frac{\Gamma \theta}{2\pi} - i \frac{\Gamma}{2\pi} \ln r = -\frac{i\Gamma}{2\pi} (\ln r + i\theta)$$

$$= -\frac{i\Gamma}{2\pi} \ln(z)$$

$$\text{at } z = z_0 \rightarrow \underline{\underline{\omega = -\frac{i\Gamma}{2\pi} \ln(z - z_0)}}$$

Stagnation point

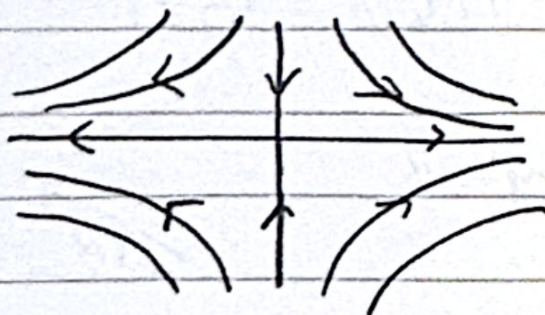
Taylor series

$$w(z) = w(z_0) + (z - z_0)w'(z_0) + \frac{1}{2}(z - z_0)^2 w''(z_0) + \dots$$

$$z = z_0 \text{ stagnation point} \rightarrow u, v = 0 \rightarrow \underline{w'(z_0) = 0}$$

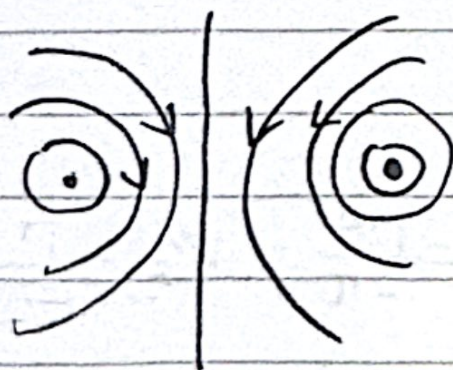
$$w \approx \frac{1}{2} \alpha z^2 \text{ near stagnation point}$$

$$u = \alpha x, \quad v = -\alpha y \quad \psi = \alpha xy$$



streamlines
(hyperbolae)

Method of Images



'image'

real

$-d$

$+d$

$$w = \frac{-i\Gamma}{2\pi} \ln(z-d) + \frac{i\Gamma}{2\pi} \ln(z+d)$$

• X-component of velocity is 0 at $x=0$ boundary.

Conformal mapping

$w(z) \rightarrow$ complex potential in z -plane.

$$Z = f(z) \text{ with inverse } z = F(Z)$$

$$\text{then } W(Z) = w\left\{\underbrace{F(Z)}_z\right\}$$

$$\text{write } Z = X + iY$$

$$W(Z) = \Phi(X, Y) + i\Psi(X, Y)$$

$$u_* = \frac{\partial \Phi}{\partial X} = \frac{\partial \Psi}{\partial Y} \quad v_* = \frac{\partial \Phi}{\partial Y} = -\frac{\partial \Psi}{\partial X}$$

$W(Z)$ & $w(z)$ take the same values at corresponding points. Streamlines & rigid boundaries are mapped accordingly.

$$\frac{dW}{dZ} = \frac{dw/dz}{dZ/dz} \rightarrow u_* - iv_* = \frac{u - iv}{f'(z)}$$

need $f'(z) \rightarrow 1$ as $z \rightarrow \infty$.

$$\arg(\delta Z_2) - \arg(\delta Z_1) = n \left[\arg(\delta z_2) - \arg(\delta z_1) \right]$$

$\uparrow$
order of the 1st non-vanishing derivative (typically 1)

Local angles are preserved ($n=1$ case).

Joukowski transformation

$$\bar{Z} = z + \frac{c^2}{z} \rightarrow z^2 + c^2 - \bar{Z}z = 0$$

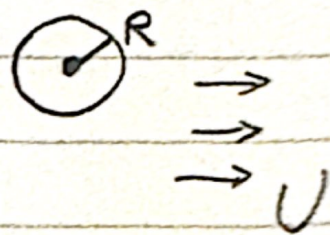
$$z = \frac{1}{2} \left(\bar{Z} + \sqrt{\bar{Z}^2 - 4c^2} \right) \quad (\text{chosen + branch})$$

$$f'(\pm c) = 0 \quad \text{but} \quad f''(\pm c) \neq 0$$

local angles around $z = \pm c$ doubled.

Flow (irrotational) past a circular cylinder

① No circulation



Ansatz (soln to Laplace's eqn)

$$\phi = \left(ar + \frac{b}{r} \right) \cos \theta$$

$$\text{b.c.} \quad \begin{cases} \phi = Ux = Ur \cos \theta & \text{at } r \rightarrow \infty \\ \frac{\partial \phi}{\partial r} = V_r = 0 & \text{at } r = R \end{cases}$$

$$\rightarrow a = U, \quad b = UR^2$$

$$\phi = U \left(r + \frac{R^2}{r} \right) \cos \theta = U \left(1 + \frac{R^2}{r^2} \right) r \cos \theta$$

$$\rightarrow \begin{cases} V_r = U \left(1 - \frac{R^2}{r^2} \right) \cos \theta \\ V_\theta = -U \left(1 + \frac{R^2}{r^2} \right) \sin \theta \end{cases}$$

$$(V_r, V_\theta = 0)$$

stagnation point at $(R, 0)$ & (R, π)

$$\text{By } \begin{cases} V_\theta = -\frac{\partial \psi}{\partial r} \\ V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} \end{cases} \rightarrow \underline{\psi = U(1 - \frac{R^2}{r^2}) r \sin \theta}$$

② With circulation

add a term of a line vortex

$$\begin{cases} \phi = U(1 + \frac{R^2}{r^2}) r \cos \theta + \frac{\Gamma \theta}{2\pi} \\ \psi = U(1 - \frac{R^2}{r^2}) r \sin \theta - \frac{\Gamma}{2\pi} \ln\left(\frac{r}{r_0}\right) \end{cases}$$

$$\begin{cases} V_r = \frac{\partial \phi}{\partial r} = \frac{U(1 - \frac{R^2}{r^2}) \cos \theta}{r} \\ V_\theta = -\frac{\partial \phi}{\partial \theta} = -U(1 + \frac{R^2}{r^2}) \sin \theta + \frac{\Gamma}{2\pi r} \end{cases}$$

$$= \frac{1}{r} \frac{\partial \psi}{\partial \theta}$$

Bernoulli's theorem:

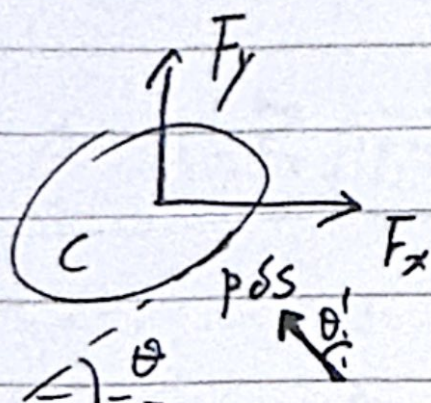
$$(P + \frac{1}{2} \rho V^2)_{r=R} = \underbrace{P_0 + \frac{1}{2} \rho U^2}_{\text{Const.}}$$

$$P = \text{Const.} - \frac{1}{2} \rho V^2$$

$$\frac{P}{\rho} = \text{Const.} - 2U^2 \sin^2 \theta + \frac{U\Gamma}{\pi R} \sin \theta \quad (\text{at } r=R)$$

$$\text{per unit length } \left\{ \begin{aligned} F_x &= \int_0^{2\pi} -P R \cos \theta d\theta = 0 \quad (\text{drag}) \\ F_y &= \int_0^{2\pi} -P R \sin \theta d\theta = -\rho U \Gamma \quad (\text{lift}) \end{aligned} \right.$$

Blasius's theorem



Principle

$$\delta F_x - i \delta F_y = -p(\sin \theta + i \cos \theta) \delta s$$
$$= -p i e^{-i\theta} \delta s$$

consider $q = \sqrt{u^2 + v^2}$

$$\frac{dw}{dz} = u - iv = q(\cos \theta - i \sin \theta) = q e^{-i\theta}$$

Bernoulli's theorem: $p = k - \frac{1}{2} \rho q^2$

$\uparrow$
const.

$$\delta F_x - i \delta F_y = \left(\frac{1}{2} \rho q^2 - k \right) i e^{-i\theta} \delta s$$
$$= \frac{1}{2} i \rho \left(\frac{dw}{dz} \right)^2 \underbrace{e^{i\theta} \delta s}_{\delta z} - k i e^{-i\theta} \delta s$$

Integrate, 2nd term vanishes, Blasius's.

$$\oint_C \delta F_x - i \delta F_y = \boxed{F_x - i F_y = \frac{1}{2} i \rho \oint_C \left(\frac{dw}{dz} \right)^2 dz}$$

Example:

flow past a circular cylinder

$$w = U \left(z + \frac{a^2}{z} \right) - \frac{i \Gamma}{2\pi} \ln z$$

$$F_x - i F_y = \frac{1}{2} i \rho \oint_C \left(U \left(1 - \frac{a^2}{z^2} \right) - \frac{i \Gamma}{2\pi z} \right)^2 dz$$

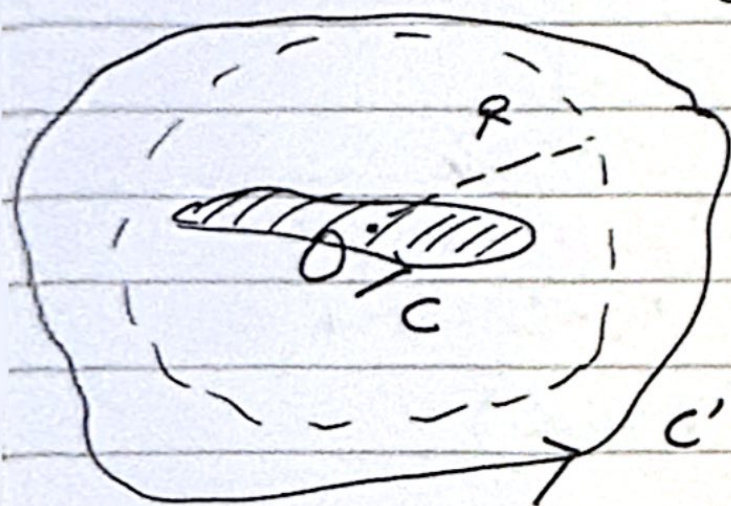
only $\bar{z}'$ term contributes,

$$F_x - iF_y = \frac{1}{2} i \rho \cdot 2\pi i \left(-i \frac{U \Gamma}{\pi} \right) = i \rho U \Gamma$$

$$\rightarrow \begin{cases} F_x = 0 \\ F_y = -\rho U \Gamma \end{cases} \quad (\text{as previously found})$$

General Kutta-Joukowski Lift Theorem

Laurant series $\frac{dw}{dz} = U + \frac{a_1}{z} + \frac{a_2}{z^2} + \dots$



$$\begin{aligned} &\rightarrow U \quad \text{Blasius's} \\ &\rightarrow F_x - iF_y \\ &\rightarrow = \frac{1}{2} i \rho \oint_{C'} \left(\frac{dw}{dz} \right)^2 dz \end{aligned}$$

only $\bar{z}'$ contributes

$$F_x - iF_y = \frac{1}{2} i \rho \cdot 2\pi i \cdot 2U a_1 = -2\pi \rho U a_1$$

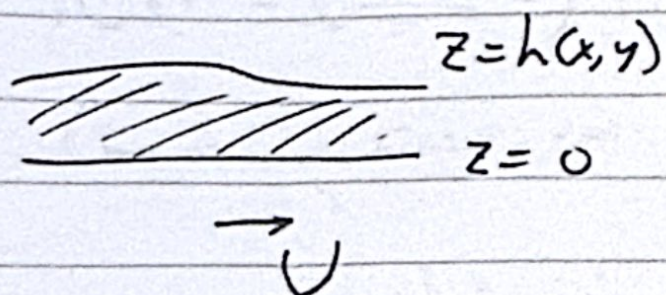
$$\text{since } \oint_{C'} \frac{dw}{dz} dz = 2\pi i a_1$$

$$\hookrightarrow \underset{(\text{Cauchy's})}{\oint_C \frac{dw}{dz} dz} = [\phi + i\psi]_C = \Gamma + i(0) = \Gamma$$

$$2\pi i a_1 = \Gamma \rightarrow 2\pi a_1 = -i\Gamma$$

$$\rightarrow F_x - iF_y = i \rho U \Gamma \rightarrow \begin{cases} F_x = 0 \\ F_y = -\rho U \Gamma \end{cases} \quad \text{Q.E.D.}$$

Thin-Film Approx. (viscous)



$h \ll L$
 ↓
 typical horizontal
 length scale
 of the flow

$$\frac{\partial^2 U}{\partial z^2} \sim \frac{U}{h^2} \quad \frac{\partial^2 U}{\partial x^2} \sim \frac{U}{L^2}$$

$$\nu \nabla^2 \underline{u} \approx \nu \frac{\partial^2 u}{\partial z^2}$$

Incompressibility gives $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

$$x, y \sim L \quad z \sim h$$

so $(\underline{u} \cdot \nabla) \underline{u} \sim \frac{U^2}{L} (1, 1, \frac{h}{L})$

$$\nu \frac{\partial^2 u}{\partial z^2} \sim \frac{\nu U}{h^2} (1, 1, \frac{h}{L})$$

can neglect $(\underline{u} \cdot \nabla) \underline{u}$ term if $\frac{U^2}{L} \ll \frac{\nu U}{h^2}$

$$\underbrace{\left(\frac{UL}{\nu} \right) \left(\frac{h}{L} \right)^2}_{Re} \ll 1$$

N-S:
$$\begin{cases} \frac{\partial p}{\partial x} = \eta \frac{\partial^2 u}{\partial z^2} \\ \frac{\partial p}{\partial y} = \eta \frac{\partial^2 v}{\partial z^2} \\ \frac{\partial p}{\partial z} = \eta \frac{\partial^2 w}{\partial z^2} \end{cases} \quad \underline{u} = (u, v, w)$$

(thin-film theory)

4

Boundary Layers

(Laminar, 2D)

($\nu \neq 0$)

Approx: ① $\left| \frac{\partial u}{\partial y} \right| \gg \left| \frac{\partial u}{\partial x} \right|$

$\delta \ll L$

thickness horizontal scale

Incompressibility $\rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

$\frac{\partial v}{\partial y} \sim \frac{U_0}{L} \rightarrow v \sim \frac{U_0 \delta}{L}, v \ll u.$

only keeps x-component of u

② $\left| \frac{\partial p}{\partial y} \right| \ll \left| \frac{\partial p}{\partial x} \right| \rightarrow \frac{\partial p}{\partial x} \approx \frac{dp}{dx}$

$(\sim \nu \frac{\partial v}{\partial y}) \quad (\sim \nu \frac{\partial u}{\partial y})$

$(\left| \frac{\partial^2 u}{\partial x^2} \right| \ll \left| \frac{\partial^2 u}{\partial y^2} \right|)$

$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \nu \frac{\partial^2 u}{\partial y^2}$

Prandtl's eqn

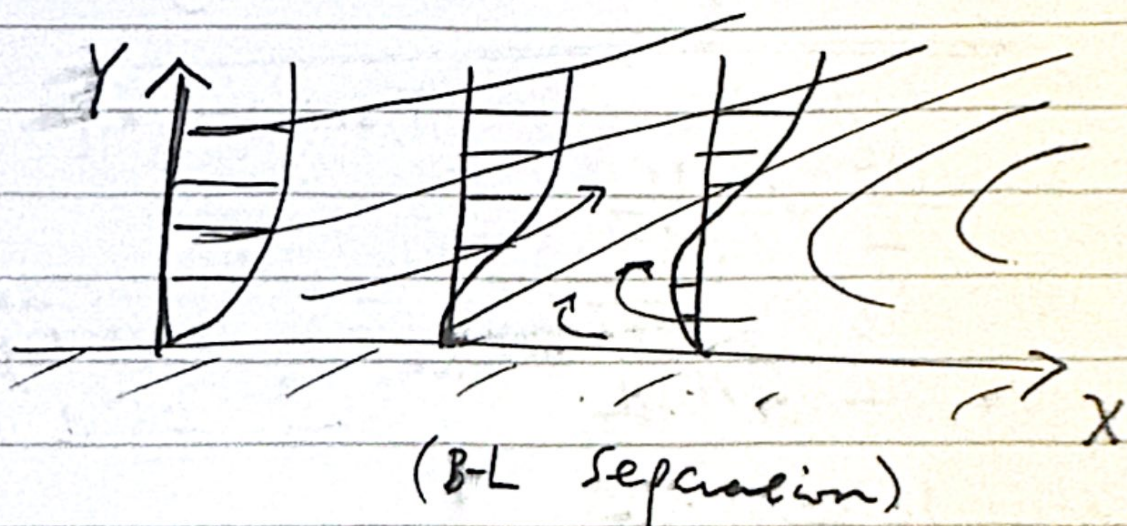
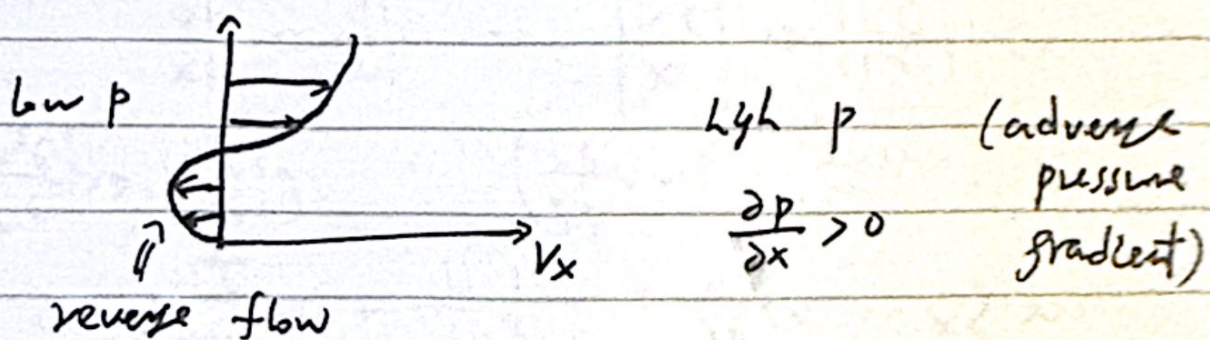
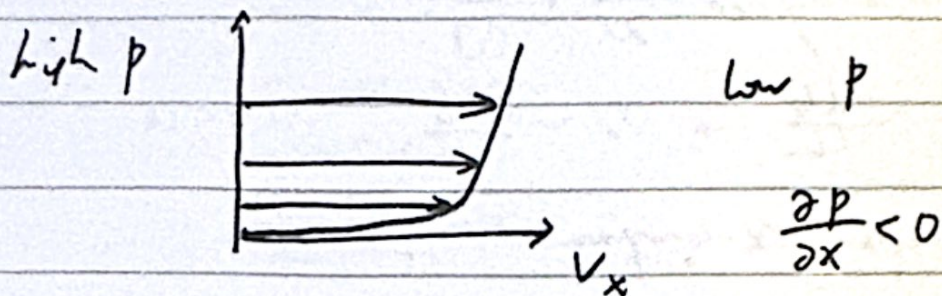
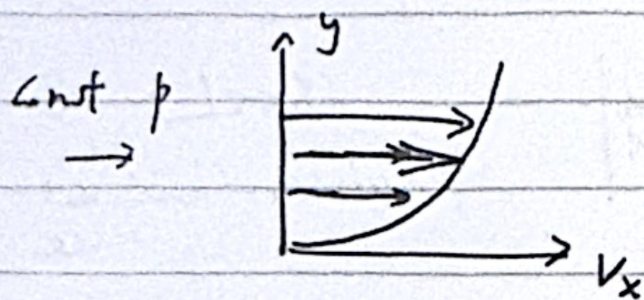
$\frac{U_0^2}{L} \sim \frac{\nu U_0}{\delta^2}$

$\rightarrow \frac{\delta}{L} \sim \sqrt{\frac{\nu}{U_0 L}} \sim (Re)^{-\frac{1}{2}} \rightarrow \text{Need large } Re, \text{ small } \nu.$

B-L Separation

- If $\frac{dp}{dx} > 0$, $U(x)$ decreases with x . The adverse pressure gradient can cause reversed flow close to the boundary.

BL diagrams



§ 5 Waves & Instability

Waves in fluids

Occurs when an equilibrium is disturbed
→ oscillating in propagating disturbances.

I Linear sound waves (small P.T.)

$$\underline{v} = \underline{v}_1 \quad p = p_0 + p_1 \quad \rho = \rho_0 + \rho_1 \quad (*)$$

Relevant eqns for sound waves:

① Euler $\rho \left(\frac{\partial \underline{v}}{\partial t} + (\underline{v} \cdot \nabla) \underline{v} \right) = -\nabla p$

② Mass Cons. $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \underline{u}) = 0$

③ Eqn of state $\frac{D}{Dt} (\rho p^{-\gamma}) = 0$

Use (*) to linearize the above eqns.

③ → $(p_0 + p_1)(\rho_0 + \rho_1)^{-\gamma} = p_0 \rho_0^{-\gamma}$

$$\left(1 + \frac{p_1}{p_0}\right) \left(1 - \frac{\gamma \rho_1}{\rho_0} + \dots\right) = 1$$

$$\frac{p_1}{p_0} = \frac{\gamma \rho_1}{\rho_0} \rightarrow p_1 = c_s^2 \rho_1$$

$$\boxed{c_s = \sqrt{\frac{\gamma p_0}{\rho_0}}} \quad \text{sound speed}$$

$$\textcircled{1} \rightarrow (\rho_0 + \rho_1) \left(\frac{\partial \underline{v}_1}{\partial t} + (\underline{v}_1 \cdot \nabla) \underline{v}_1 \right) = -\nabla(p_0 + p_1)$$

$\nabla p_0 = 0$, ignore higher orders

$$\underline{\rho_0 \frac{\partial \underline{v}_1}{\partial t} = -\nabla p_1}$$

$$\textcircled{2} \rightarrow \underline{\frac{\partial \rho_1}{\partial t} + \rho_0 \nabla \cdot \underline{v}_1 = 0}$$

$$\text{Thus } \rho_0 \frac{\partial}{\partial t} (\nabla \cdot \underline{v}_1) = \frac{\partial^2 \rho_1}{\partial t^2} = -\nabla^2 p_1$$

$$\boxed{\frac{\partial^2 p_1}{\partial t^2} = c_s^2 \nabla^2 p_1}$$

$$\underline{\text{1D}} \quad \frac{\partial^2 p_1}{\partial t^2} = c_s^2 \frac{\partial^2 p_1}{\partial x^2}$$

$$\underline{p_1(x, t) = f(x - c_s t) + g(x + c_s t)}$$

spherical symmetric

$$\frac{\partial^2 p_1}{\partial t^2} = c_s^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial p_1}{\partial r} \right)$$

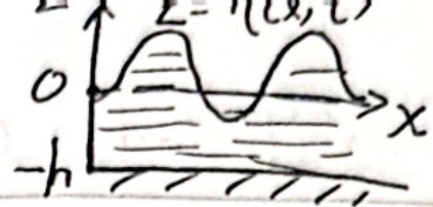
$$\text{simply} \rightarrow \frac{\partial^2}{\partial t^2} (r p_1) = c_s^2 \frac{\partial^2}{\partial r^2} (r p_1)$$

$$\underline{p_1 = \frac{1}{r} [f(r - c_s t) + g(r + c_s t)]}$$

no waves from infinity (radiation condition)

$$\rightarrow p_1 = \frac{1}{r} F(r - c_s t) \quad \text{only}$$

II Surface gravity waves



inviscid & incompressible, $\nabla \cdot \mathbf{v} = 0$

Boundary conditions

surface at $z = \eta(x, t)$

$$\mathbf{v} = \mathbf{v}_0 + \mathbf{v}_1 \quad p = p_0 + p_1$$

Euler's eqn: $\rho_0 \frac{\partial \mathbf{v}_1}{\partial t} = -\nabla(p_0 + p_1) - \rho_0 g \hat{z}$

$$\mathbf{v}_1 = \nabla \phi \rightarrow \nabla \left(\rho_0 \frac{\partial \phi}{\partial t} + p_1 \right) = \underbrace{-\nabla p_0 - \rho_0 g \hat{z}}_0$$

~~$p_1 = p_0(\eta) = p_1$~~ (0 too)

$$(p_1 - p_{atm}) \delta x = -\gamma \left(\left. \frac{\partial \eta}{\partial x} \right|_{x+\delta x} - \left. \frac{\partial \eta}{\partial x} \right|_x \right) \approx -\gamma \frac{\partial^2 \eta}{\partial x^2} \delta x$$

$$p_1 = p_{atm} - \gamma \frac{\partial^2 \eta}{\partial x^2} = \rho_0 g \eta - \gamma \frac{\partial^2 \eta}{\partial x^2}$$

$$\text{so } \frac{\partial \phi}{\partial t} = -g\eta + \frac{\gamma}{\rho_0} \frac{\partial^2 \eta}{\partial x^2}$$

at $z=0$ dynamical b.c.

$$v_z = \frac{dz}{dt} = \frac{\partial \eta}{\partial t}$$

(A)

$$\frac{\partial \phi}{\partial z} = \frac{\partial \eta}{\partial t}$$

at $z=0$

kinematic b.c.

(B)

$$\frac{\partial \phi}{\partial z} = 0$$

at $z=-h$

no penetration
(at rigid boundary)

(C)

Dispersion (surface waves)

Consider ansatz $\eta(x, t) = A \cos(kx - \omega t)$

$$(z=0) \quad \textcircled{A} \rightarrow \frac{\partial \phi}{\partial t} = -\left(g + \frac{\gamma k^2}{\rho_0}\right) A \cos(kx - \omega t) \quad \textcircled{A'}$$

$$(z=0) \quad \textcircled{B} \rightarrow \frac{\partial \phi}{\partial z} = A\omega \sin(kx - \omega t) \quad \textcircled{B'}$$

$$\phi(x, z, t) = \underline{f(z)} \sin(kx - \omega t)$$

↓ into Laplace's eqn

$$f''(z) - k^2 f(z) = 0$$

$$f(z) = C_1 e^{kz} + C_2 e^{-kz}$$

$$\text{At } -h, \frac{\partial \phi}{\partial z} = 0 \rightarrow \underline{C_2 = C_1}$$

$$\textcircled{C} \quad C_1 e^{-kh} - C_2 e^{kh} = 0$$

$$\rightarrow \underline{f(z) = C \cosh(k(z+h))}$$

$$\phi(x, z, t) = C \cosh(k(z+h)) \sin(kx - \omega t)$$

$$\textcircled{A'} \rightarrow C(-\omega) \underset{\rightarrow 0}{\cosh(k(z+h))} = -\left(g + \frac{\gamma k^2}{\rho_0}\right) A$$

$$C = \frac{A}{\omega \cosh(kh)} \left(g + \frac{\gamma k^2}{\rho_0}\right)$$

$$\textcircled{B'} \rightarrow Ck \sinh(kh) = A\omega \rightarrow A = \frac{Ck}{\omega} \sinh(kh)$$

$$\text{Thus } 1 = \frac{k}{\omega^2} \tanh(kh) \left(g + \frac{\gamma k^2}{\rho_0}\right)$$

Dispersion relation:

$$\omega^2 = \left(gk + \frac{\gamma k^3}{\rho_0} \right) \tanh(kh)$$

Velocity potential:

$$\phi = \frac{\omega A}{k} \sin(kx - \omega t) \frac{\cosh(k(z+h))}{\sinh(kh)}$$

Phase speed:

$$V_\phi = \frac{\omega}{k} = \sqrt{\left(\frac{g}{k} + \frac{\gamma k}{\rho_0} \right) \tanh(kh)}$$

Gravity-Capillary waves

(ripples)

$$kh \gg 1 \quad (\text{small wavelength})$$

$$\omega^2 = gk \left(1 + \frac{\gamma k^2}{g\rho_0} \right) = \underline{gk \left(1 + \lambda_c^2 k^2 \right)}$$

$$\left(\omega^2 \approx \frac{\gamma k^3}{\rho_0} \right)$$

$$\text{capillary length } \lambda_c = \sqrt{\frac{\gamma}{g\rho_0}}$$

$$\hookrightarrow \begin{cases} V_\phi = \frac{\omega}{k} = \sqrt{\frac{\gamma k}{\rho_0}} \end{cases}$$

$$\begin{cases} V_g = \frac{d\omega}{dk} = \frac{3}{2} \sqrt{\frac{\gamma k}{\rho_0}} = \frac{3}{2} V_\phi \end{cases}$$

III

Internal gravity waves

Buoyancy freq

$$\text{EoM: } \rho_0 \frac{d^2(\delta z)}{dt^2} = g \left(\frac{d\rho_0}{dz} \right) \delta z$$

$$\hookrightarrow N = \sqrt{-\frac{g}{\rho_0} \frac{d\rho}{dz}}$$

Euler Egn

$$\rho \left(\frac{\partial \underline{v}}{\partial t} + (\underline{v} \cdot \nabla) \underline{v} \right) = -\nabla p - \rho g \hat{z}$$

Incompress.

$$\nabla \cdot \underline{v} = 0$$

Mass Cons

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + (\underline{v} \cdot \nabla) \rho = 0$$

Linearize again by $\begin{cases} \underline{v} = \underline{v}_1 \\ p = p_0 + p_1 \\ \rho = \rho_0 + \rho_1 \end{cases}$

$$\rho_0 \frac{\partial v_{1,x}}{\partial t} = -\frac{\partial p_1}{\partial x}$$

$$\rho_0 \frac{\partial v_{1,z}}{\partial t} = -\frac{\partial p_1}{\partial z} - \rho_1 g$$

$$\frac{\partial v_{1,x}}{\partial x} + \frac{\partial v_{1,z}}{\partial z} = 0$$

$$\frac{\partial \rho_1}{\partial t} + v_{1,z} \frac{d\rho_0}{dz} = 0$$

variables:

$$(v_{1,x}, v_{1,z}, p_1, \rho_1)$$

Assume all varying with $\sim e^{i(k_x x + k_z z - \omega t)}$

$$\begin{cases} \omega \rho_0 \hat{v}_x - k_x \hat{p} = 0 \\ i\omega \rho_0 \hat{v}_z - i k_z \hat{p} - g \hat{\rho} = 0 \\ k_x \hat{v}_x + k_z \hat{v}_z = 0 \\ \frac{d\rho_0}{dz} \hat{v}_z - i\omega \hat{\rho} = 0 \end{cases} \quad \begin{array}{l} \text{for non-trivial,} \\ \text{solve} \\ \text{by } \det(M) = 0 \end{array}$$

Dispersion (internal gravity waves)

$$\omega^2 = \frac{N^2 k_x^2}{k_x^2 + k_z^2} \rightarrow \underline{\omega} = N \sin \theta = \frac{N k_x}{\sqrt{k_x^2 + k_z^2}}$$

$$\underline{V_\phi} = \frac{N k_x}{k^2} \frac{\underline{k}}{k}$$

$$\underline{k} = (k_x, k_z)$$

$$k = \sqrt{k_x^2 + k_z^2}$$

$$\underline{V_g} = \frac{\partial \omega}{\partial k_x} \hat{x} + \frac{\partial \omega}{\partial k_z} \hat{z}$$

$$\begin{aligned} \frac{\partial \omega}{\partial k_x} &= \frac{\partial}{\partial k_x} \left(\frac{N k_x}{\sqrt{k_x^2 + k_z^2}} \right) = N \left[\frac{k_x^2}{k_x^2 + k_z^2} - \frac{k_x^2}{\sqrt{k_x^2 + k_z^2}^3} \right] \\ &= \frac{N k_z^2}{k^3} \end{aligned}$$

$$\frac{\partial \omega}{\partial k_z} = - \frac{N k_x k_z}{k^3}$$

$$\underline{V_g} = \frac{N k_z}{k^3} (k_z \hat{x} - k_x \hat{z})$$

Instability

Similar means as in linearisation
& boundary cond. calc. for dispersion

→ Unstable if $\text{Im}(\omega) = \sigma > 0$
exp growth

Turbulence

$$\underline{v} = \underset{\substack{\uparrow \\ \text{mean}}}{V} + \underset{\substack{\uparrow \\ \text{fluctuation}}}{v'} \quad (\langle \underline{v} \rangle = V)$$

$$\langle v_i v_j \rangle = \langle V_i V_j \rangle + \underbrace{V_i \langle v_j' \rangle}_0 + \underbrace{V_j \langle v_i' \rangle}_0 + \langle v_i' v_j' \rangle$$

$$\langle v_i v_j \rangle = \underbrace{\langle V_i V_j \rangle}_{\text{mean transport}} + \underbrace{\langle v_i' v_j' \rangle}_{\text{eddy flux}}$$

incompressibility $\nabla \cdot \underline{v} = 0$

$$\rightarrow \langle \nabla \cdot \underline{v} \rangle = \nabla \cdot \langle \underline{v} \rangle = 0$$

$$\underline{\nabla} \cdot \underline{V} = 0$$

$$\text{so } \nabla \cdot \underline{v}' = \nabla \cdot \underline{v} - \nabla \cdot \underline{V} = 0$$

N-S index form:

$$\frac{\partial}{\partial t} (\rho v_i) + \frac{\partial}{\partial x_j} (\rho v_i v_j) = - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) \right)$$

take $\langle \dots \rangle$:

$$\begin{aligned} & \frac{\partial}{\partial t} \rho V_i + \frac{\partial}{\partial x_j} (\rho V_i V_j) \\ &= - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\eta \left(\frac{\partial V_i}{\partial x_j} + \frac{\partial V_j}{\partial x_i} \right) \right) - \underbrace{\frac{\partial}{\partial x_j} (\tau_{ij})}_{\frac{\partial}{\partial x_j} (\rho \langle v_i' v_j' \rangle)} \end{aligned}$$

Reynolds stress

$$\tau_{ij} = -\rho \langle v_i v_j' \rangle$$

→ Correlation between components of fluctuation carries momentum transport too.

Mixing length

in kinetic theory

$$\nu \propto u \lambda$$

↑ ↑
thermal m.f.p.
vel

By analogy

in turbulent flow →

$$\tau_{ij} \sim \rho \nu_T$$

↓
turbulent
eddy viscosity

$$\boxed{\nu_T \propto V_T \Lambda}$$

↓
turbulent
velocity
scale

← mixing length
(scale of distance before
decorrelation in system)